

PROJECT TRICORDER MK I

TECHNICAL SPECIFICATION: TERMUX MULTI-SERVER & TRICORDER-CORE TUPLE LAWS

Executive Architectural Summary

Project Tricorder Mk I represents an offline-first, edge-AI multi-sensor field instrument designed by **Jim Ames, Claude, and Grok** at **N2NHU Labs** (Newburgh, NY). Grounded in the **Multi-Tronic Operating Realm (M-TOR)** paradigm, the architecture rejects cloud dependencies and monolithic app wrappers. Instead, it operates as a hybrid microkernel where high-frequency hardware sensors and native TFLite classifiers reside in privileged Android Kotlin space (the *Visceral Spine*), while heavy generative language, vision, image, and voice models run as isolated user-space microservices inside **Termux** on loopback ports. Deterministic intent resolution is enforced by the zero-LLM algebraic tuple engine in tricorder-core.

The M-TOR Design Mandate: *Point. Name. Speak. Store. Show.*

1. Nothing important is hardcoded (data-driven models, routes, and providers).
2. API-first architecture (the UI is a thin renderer of bus events).
3. Event-driven pub/sub communication (EventBus decouples UI, organs, and engines).
4. Algebraic intent routing (deterministic tuple vector matching without LLM latency at the door).
5. Extensible by construction (context providers follow standardized row molecules).

1. Termux Multi-Server Port Configuration

To maintain real-time responsiveness without crashing the Android OS or stalling the UI thread, Project Tricorder isolates four generative weights into separate local HTTP processes inside Termux. All four ports remain continuously warm on loopback (127.0.0.1) on Snapdragon 888 reference hardware (Samsung Galaxy Z Fold 3, 12 GB RAM budget).

Port	Role / Organ	Binary / Daemon	Model Weights / Engine	Protocol / Endpoint
:8080	Inquiry / Chat	llama-server	Gemma 3 4B Instruct Q4_K_M (2048/8192 ctx, 3 threads)	openai_chat_vision/v1/chat/completions
:8081	Sensor & Sentry Vision	llama-server	SmolVLM2-500M-Video-Instruct-Q8_0 + mmproj-f16	openai_chat_vision/v1/chat/completions
:8188	Imagine (Pictograms)	sd-server	SD 2.1 Turbo Q4_0 (--cfg-scale 1, 4 steps)	openai_images/v1/images/generations

Port	Role / Organ	Binary / Daemon	Model Weights / Engine	Protocol / Endpoint
:5000	Voice / Measured TTS	python3 server.py	Kokoro-v1.0 ONNX (82M, fp32) + voices-v1.0.bin	openai_audio_spe ech /v1/audio/speech
:8022	Remote Maintenance	sshd	Termux OpenSSH Daemon	SSH Protocol

Termux Stack Lifecycle & Launcher Script

The Termux stack is managed by the shell script `~/bin/tricorder`. Processes are started via `setsid nohup` with PID tracking in `~/tricorder-pids` and logs routed to `~/tricorder-logs`. Background execution is protected using `termux-wake-lock` and setting Android battery optimization for Termux to **Unrestricted**.

```
#!/data/data/com.termux/files/usr/bin/bash
# Project Tricorder Mk I – Daemon Lifecycle Manager (~/bin/tricorder)
cmd="{1:-status}"

case "$cmd" in
start)
termux-wake-lock
sshd
start_one chat $LLAMA -m $GEMMA --host 127.0.0.1 --port 8080 -c 2048 -t 3
start_one vision $LLAMA -m $SMOL --mmproj $MMPROJ --host 127.0.0.1 --port 8081 -c 1024 -t 2
start_one image $SD -m $SDMODEL --listen-ip 127.0.0.1 --listen-port 8188 --cfg-scale 1
start_one tts bash -lc "cd $KOKORO_DIR && python3 server.py"
;;
stop)
stop_one tts; stop_one image; stop_one vision; stop_one chat;
;;
status)
status_one chat 8080; status_one vision 8081; status_one image 8188; status_one tts 5000;
;;
esac
```

2. The tricorder-core Algebraic Intent Engine

The Node package `tricorder-core` serves as the software constitution. Rather than relying on non-deterministic LLM evaluation at the front door—which introduces system lag, consumes tokens, and causes unpredictable behavior—operator intent is projected onto a 4-element algebraic tuple:

Tuple Definition: `I = (Mode, InputChannel, TargetModality, Action)`

- **Mode** — Current top-level state (e.g., CHATTING, SCANNING, GUARDING, IDLE).
- **InputChannel** — Origin of stimulus (e.g., AUDIO_IN, CAMERA, TEXT, TIMER).
- **TargetModality** — Desired output type (e.g., TEXT, IMAGE_GEN, VISION, MOTION, AUTO).
- **Action** — Execution directive (e.g., EXECUTE, TOGGLE, LOOP, STOP).

Keyword Override Hints & First-Match Resolution

Control buttons seed a baseline tuple (e.g., Button I seeds [CHATTING, AUDIO_IN, TEXT, TOGGLE]). Before rule evaluation, free-text input (speech transcript or typed query) is scanned against intent_keyword_hints defined in tricorder.config.json. For example, keywords like 'imagine' or 'show me an image of' dynamically bump TargetModality to IMAGE_GEN. The TupleEngine then evaluates the ordered tuple_rules array — **first matching rule wins**.

Match Pattern	Primary Route (route_to)	Fallback Target	Execution Characteristics
{ mode: CHATTING, target_hint: IMAGE_GEN }	models.image_generation (:8188 SD Turbo)	models.chat_vision (:8080 Gemma)	Inquiry prompt asking for image. Falls back to chat description if SD server unavailable.
{ mode: CHATTING, target_hint: * }	models.chat_vision (:8080 Gemma)	None	Standard conversational inquiry. Streamed tokens to CRT + chunked Kokoro TTS.
{ mode: SCANNING }	models.vision_fast (:8081 SmolVLM)	None	Sensor Scan Loop: Captures frame + calls VLM every 3500ms unconditionally.
{ mode: GUARDING }	sensor.motion_delta (In-process pixel-diff)	Escalates to: models.vision_fast	Sentry Guard Loop: Polls local motion every 750ms. Escalates to VLM only if motion >= 12.5%.

3. Core Subsystem Vocabulary & Sensor Drivers

Module File	Design Spec Role	Core Responsibilities & Operations
eventBus.mjs	Pub/Sub Spine	Central pub/sub bus. UI, providers, and engines never touch directly. Emits Events.BUTTON_PRESS, TUPLE_RESOLVED, SENSOR_DESCRIPTION, ALERT_RAISED, etc.
configLoader.mjs	Configuration Host	Loads, validates, and hot-reloads tricorder.config.json. Emits config.error on malformed JSON without crashing.
tupleEngine.mjs	Algebraic Router	Transforms button seeds & transcripts into tuples, applies keyword hints, and resolves matching route rules.
mcpClient.mjs	MCP Tool Client	Resolves dotted route paths (e.g. models.chat_vision), dispatches HTTP requests or local handlers, and executes fallbacks seamlessly.
payloadBuilder.mjs	Protocol Translator	Translates generic IntentContext ({ text, images }) into backend-specific body shapes (e.g., openai_chat_vision vs openai_images).
sensorLoopDriver.mjs	Loop Controller	Drives repeating Scan Loops (Button II) and Guard Loops (Button III). Manages motion escalation thresholds and prevents VLM flooding.

Module File	Design Spec Role	Core Responsibilities & Operations
contextProviderPoller.mjs	Context Poller	Generic repeating fetch-and-normalize driver for environmental/civic feeds (e.g., noaaWeatherAlerts.mjs, OpenSky).

Scan Loops vs. Guard Loops (Sensor & Sentry)

The SensorLoopDriver implements two distinct loop paradigms based on the structure of the resolved rule:

- **Scan Loop (Button II - Sensor Mode):** Captures a camera frame and POSTs to models.vision_fast (:8081) unconditionally every loop_interval_ms (default 3500ms). Emits SENSOR_DESCRIPTION.
- **Guard Loop (Button III - Sentry Mode):** Polls in-process local frame-differencing (sensor.motion_delta) every motion_poll_interval_ms (750ms). No network or VLM calls occur while pixel variance remains below escalation_threshold_pct (12.5%). Once crossed, it captures a frame, escalates to SmolVLM2 once, and emits ALERT_RAISED.

4. Grounded Engineering Boundaries & Safety Warnings

CRITICAL NAVIGATION & SAFETY WARNING:

Project Tricorder Mk I is an experimental field instrument and edge-AI research platform. It is **NOT certified** by the FAA, NASA, EASA, or ICAO, and does not comply with DO-178C airborne software standards.

- **Horizon & Navigation:** Attitude Direction Indicator (ADI), heading, G-meter, GPS speed, and barometric altitude proxies are field novelties and **MUST NEVER** be used to fly, land, or navigate any craft.
- **Sentry & Watch:** Weapon and gunfire detection lines are keyword-matching captions and YAMNet audio labels — **NOT** a certified weapons detector or 911 replacement.
- **Medical & Memory Alpha:** Medical queries return encyclopedia briefings (WikiMed extracts), **NOT** clinical medical advice. Memory Alpha searches Simple English Wikipedia extracts, **NOT** fictional lore databases.

Document Control: Technical Manual Rev 0.6.71	Authors: Jim Ames, Claude & Grok (N2NHU Labs)	Reference Hardware: Galaxy Z Fold 3 (Snapdragon 888)
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